Lecture Notes on Multivariable Calculus

Notes written by Barbara Niethammer and Andrew Dancer

Lecturer Jan Kristensen

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1 Introduction

In this course we shall extend notions of differential calculus from functions of one variable to more general functions

$$f\colon \mathbb{R}^n \to \mathbb{R}^m. \tag{1.1}$$

In other words functions $f = (f_1, \ldots, f_m)$ with m components, all depending on n variables $x_1, \ldots, x_n$. In fact, we shall take it one step further and consider maps

$$f\colon X \to Y$$

where X and Y are normed vector spaces over $\mathbb{R}$. In this course we restrict attention to the case where X and Y are *finite* dimensional, and so by a choice of bases in X and in Y we reduce this more general looking case to (**??**). It is however often convenient to have this *invariant* definition.

The questions we will be interested in studying include the following.

- If $n = m$, under what circumstances is f invertible?
- If $n \geq m$, when does the equation $f(x, y) = 0$, with $x \in \mathbb{R}^{n-m}$ and $y \in \mathbb{R}^m$, implicitly determine y as a function of x ?
- Is the zero locus $f^{-1}(0)$ a smooth subset of $\mathbb{R}^n$ in a suitable sense, for example a smooth surface in $\mathbb{R}^3$?

The first major new idea is to define the derivative at a point as a linear map, which we can think of as giving a first-order approximation to the behaviour of the function near that point. A key theme will be that, subject to suitable nondegeneracy assumptions, the derivative at a point will give qualitative information about the function on a neighbourhood of the point. In particular, the Inverse Function Theorem will tell us that invertibility of the derivative at a point (as a linear map) will actually guarantee local invertibility of the function in a neighbourhood.

The results of this course are foundational for much of mathematics and the notion of a *smooth manifold* is in particular central to mathematical physics and geometry. A smooth submanifold in $\mathbb{R}^n$ is, intuitively, a generalisation to higher dimensions of the notion of a smooth surface in $\mathbb{R}^3$. Among many other things we shall use our theorems to obtain a criterion for when the locus defined by a system of nonlinear equations is a manifold. Manifolds are the setting for much of higher-dimensional geometry and mathematical physics and in fact the concepts of differential (and integral) calculus that we study in this course can be developed on general manifolds. The Part B course Geometry of Surfaces and the Part C course Differentiable Manifolds develop these ideas further.

2 Differentiation of functions of several variables

2.1 Introduction

In this chapter we will extend the concept of differentiability of a function of one variable to the case of a function of several variables. We first recall the definitions for a function of one variable.

Differentiability of a function of one variable

Let $I \subseteq \mathbb{R}$ be an open interval. A function $f\colon I \subseteq \mathbb{R} \to \mathbb{R}$ is differentiable at $x \in I$ if

$$f'(x) := \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

exists. Equivalently we can say that f is differentiable in $x \in I$ if there exists a linear map* $L\colon \mathbb{R} \to \mathbb{R}$ such that

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - Lh}{h} = 0. \tag{2.1}$$

In this case, L is given by $L : h \mapsto f'(x) \cdot h$.

Another way of writing (**??**) is

$$f(x+h) - f(x) - Lh = R_f(h) \qquad \text{with } R_f(h) = o(|h|), \quad \text{i.e.} \quad \lim_{h\to 0} \frac{R_f(h)}{|h|} = 0. \tag{2.2}$$

This definition is more suitable for the multivariable case, where h is now a vector, so it does not make sense to divide by h.

Differentiability of a vector-valued function of one variable

Completely analogously we define the derivative of a vector-valued function of one variable. More precisely, if $f\colon I \subseteq \mathbb{R} \to \mathbb{R}^m, m > 1$, with components $f_1, \ldots, f_m$, we say that f is differentiable at $x \in I$ if

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

exists.

*Here and in what follows, we will often write Lh instead of $L(h)$ if L is a linear map.

It is easily seen that f is differentiable at $x \in I$ if and only if $f_i \colon I \subseteq \mathbb{R} \to \mathbb{R}$ is differentiable in $x \in I$ for all $i = 1, \ldots, m$. Also, f is differentiable in $x \in I$ if and only if there exists a linear map $L \colon \mathbb{R} \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{f(x+h) - f(x) - Lh}{h} = 0.$$

How can we now generalize the concept of differentiability to functions of several variables, say for a function $f \colon \Omega \subseteq \mathbb{R}^2 \to \mathbb{R}$, $f = f(x, y)$? A natural idea is to freeze one variable, say y, define $g(x) = f(x, y)$ and check whether g is differentiable at x. This will lead to the notion of partial derivatives and most of you have seen this already in lectures in the first year, e.g. in Calculus. However, we will see that the concept of partial derivatives alone is not completely satisfactory. For example we will see that the existence of partial derivatives does not guarantee that the function itself is continuous (as it is the case for a function of one variable).

The notion of the (total) derivative for functions of several variables will not have this deficiency. It is based on a generalisation of the formulation in (**??**). In order to do that we will need a suitable norm (length function) on $\mathbb{R}^n$. You may have learned already, e.g. in Topology, that all norms on $\mathbb{R}^n$ are equivalent, and hence properties of sets, such as openness or boundedness, and of functions, such as continuity, do not depend on the choice of the norm.

In the sequel we will always use the *Euclidean norm* on $\mathbb{R}^n$ and denote it by $|\cdot|$. More precisely, for $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we denote

$$|x| = \sqrt{x_1^2 + \ldots x_n^2}\,.$$

You may check yourself that this defines a norm (and hence a metric). For the proof of the triangle inequality you will need to use the Cauchy-Schwarz inequality.

We shall also use the matrix (Hilbert-Schmidt) norm

$$\|C\| = \left(\sum_{i,j=1}^{n} C_{ij}^2 \right)^{\frac{1}{2}}$$

on the space of $n \times n$ real matrices. We have the following useful inequality:

$$|Ch| = \Big(\sum_i \Big(\sum_j C_{ji} h_j \Big)^2 \Big)^{\frac{1}{2}} \leq \Big(\sum_i \Big(\sum_j C_{ij}^2 \Big) \Big(\sum_j h_j^2 \Big) \Big)^{\frac{1}{2}} = |h|\, \|C\|.$$

We shall also occasionally use the fact that

$$\|AB\| \leq \|A\| \|B\|.$$

2.2 Partial derivatives

We are going to consider[†] functions $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$. Here and in what follows we always assume that Ω is an open and connected subset of $\mathbb{R}^n$ (a domain).

We first consider a few examples of such functions.

Example 2.2.1. a) The function $f\colon \mathbb{R}^n \to \mathbb{R}$ given by

$$x \mapsto |x| = \left(\sum_{i=1}^{n} x_i^2\right)^{\frac{1}{2}}.$$

b) The function $g : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$g\colon (x, y, z) \mapsto \begin{pmatrix} x^3 + y^3 + z^3 - 7 \\ xy + yz + zx + 2 \end{pmatrix}.$$

c) The electric field in a vacuum induced by a point charge q in a point $x_0 \in \mathbb{R}^3$ is given by

$$f\colon \mathbb{R}^3 \setminus \{x_0\} \to \mathbb{R}^3, \qquad f(x) = q\frac{x - x_0}{|x - x_0|^3}.$$

d) $f\colon \mathbb{C} \to \mathbb{C}, f(z) = e^z$ has – with the usual identifications – the real representation $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}.$$

e) $f\colon \mathbb{R}^{n^2} \to \mathbb{R}^{n^2}$ where we identify $\mathbb{R}^{n^2}$ with $M_{n\times n}(\mathbb{R})$, the vector space of $n \times n$ matrices, and define f by $f : A \mapsto A^2$.

We shall sometimes use the concepts of *graphs* and *level sets*. Let $f\colon \Omega \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a function. Then the graph of f, given by

$$\Gamma_f = \{(x, y) \in \Omega \times \mathbb{R} \mid y = f(x)\},$$

is usually a surface in $\mathbb{R}^3$. Its level set at level $c \in \mathbb{R}$ is

$$N_f(c) = \{x \in \Omega \mid f(x) = c\},$$

which is usually a curve in Ω.

The concepts of graphs and level sets generalise in an obvious way to functions $f\colon \Omega \to \mathbb{R}$ defined on domains Ω in $\mathbb{R}^n$ and beyond.

[†]We will use the shorthand $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$' to mean that "$\Omega$ is a domain in $\mathbb{R}^n$ and $f\colon \Omega \to \mathbb{R}^m$ is a function".

Definition 2.2.2. (Partial derivative) Let $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$. We say that the i-th partial derivative of f in $x \in \Omega$ exists, if

$$\frac{\partial f}{\partial x_i}(x) = \lim_{t\to 0} \frac{f(x+te_i) - f(x)}{t}$$

exists, where e_i is the i-th unit vector. In other words, the i-th partial derivative is the derivative of $g(t) = f(x+te_i)$ at $t = 0$.

Other common notations for the i-th partial derivative of f at x are

$$\partial_i f(x), \quad D_i f(x), \quad \partial_{x_i} f(x), \quad f_{x_i}(x) \quad \text{or} \quad f_{,i}(x).$$

We will mostly use $\partial_i f(x)$. If $f\colon \Omega \subseteq \mathbb{R}^2 \to \mathbb{R}$ we often write $f = f(x,y)$ and denote the partial derivatives by $\partial_x f$ and $\partial_y f$ respectively.

Example 2.2.3. a) Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be given by $f(x) = |x|$. Then for $x \neq 0$ we have

$$\frac{1}{t}\big(|x+te_i| - |x|\big) = \frac{1}{t}\frac{|x+te_i|^2 - |x|^2}{|x+te_i| + |x|} = \frac{1}{t}\frac{2tx_i + t^2}{|x+te_i| + |x|} = \frac{2x_i + t}{|x+te_i| + |x|} \to \frac{x_i}{|x|}$$

as $t \to 0$. Hence, for $x \neq 0$, the function f has partial derivatives, given by $\partial_i f(x) = \frac{x_i}{|x|}$. Notice that no partial derivative of f exists at $x = 0$.

b) Let $f(x) = g(r)$ with $r(x) = |x|$ and differentiable $g\colon [0,\infty) \to \mathbb{R}$. Then, for $x \neq 0$, by the Chain Rule from Prelims Analysis 2, we find

$$\partial_i f(x) = g'(r)\partial_i r(x) = g'(r)\frac{x_i}{|x|} = \frac{g'(r)}{r}x_i\,.$$

The following example shows that, surprisingly, functions whose partial derivatives all exist are in general not continuous.

Example 2.2.4. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$

$$f(x,y) = \begin{cases} \dfrac{xy}{(x^2+y^2)^2} & \text{for } (x,y) \neq (0,0), \\[2ex] 0 & \text{for } (x,y) = (0,0). \end{cases}$$

- f has partial derivatives on $\mathbb{R}^2 \setminus \{(0,0)\}$ with

  $$\partial_x f(x,y) = \frac{y}{(x^2+y^2)^2} - \frac{4x^2y}{(x^2+y^2)^3}$$

  and a similar expression for $\partial_y f(x,y)$.

- f has partial derivatives at 0, since f is identically zero on the x and y axes. Explicitly, for all $t \neq 0$ we have

  $$\frac{f((0,0)+t(1,0)) - f(0,0)}{t} = \frac{f(t,0) - f(0,0)}{t} = \frac{0-0}{t} = 0$$

  and thus $\partial_x f(0) = 0$; similarly $\partial_y f(0) = 0$.

- **But:** f is not continuous at $(x, y) = (0, 0)$. To see this, consider the behaviour of the function on the line $\{(t, t) : t \in \mathbb{R}\}$. On this line, $f(t, t) = \frac{1}{4t^2}$ which tends to ∞ as $t \to 0$.

This shows that existence of partial derivatives is *not* the correct notion of differentiability in higher dimensions. We shall see later that the correct higher dimensional version of differentiability, using the ideas concerning linear maps from the beginning of this chapter, *will* imply continuity of the function. We shall also see that functions with *continuous* partial derivatives are differentiable in this correct sense and hence are also continuous. In our example above, the partial derivatives $\partial_i f(x)$ are not continuous at $x = 0$.

Before we define differentiability, we shall make some more remarks about partial derivatives.

The partial derivative is a special case of the *directional derivative* which we will now define.

Definition 2.2.5. (Directional derivative) Suppose that $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ and let $v \in \mathbb{R}^n \backslash \{0\}$. If

$$\partial_v f(x) = \lim_{t \to 0} \frac{f(x + tv) - f(x)}{t}$$

exists, we call it the *directional derivative* of f in direction v at the point $x \in \Omega$.

Observe that if v is one of the unit coordinate vectors e_i, then we recover the notion of partial derivative.

Example 2.2.6. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be given by $f(x) = |x|$ and let $x, v \in \mathbb{R}^n \setminus \{0\}$. Then

$$\begin{aligned} \partial_v f(x) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} |x + tv| = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \Big(\sum_{i=1}^n |x_i + tv_i|^2 \Big)^{\frac{1}{2}} \\ &= \frac{1}{2|x|} \sum_{i=1}^n 2x_i v_i = \sum_{i=1}^n \frac{x_i}{|x|} v_i = \Big\langle \frac{x}{|x|}, v \Big\rangle , \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ denotes the usual scalar product in $\mathbb{R}^n$. If $v = e_i$ we recover the formula $\partial_i |x| = \frac{x_i}{|x|}$ from Example **??**.

Definition 2.2.7. (Gradient) Let $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ and assume that all partial derivatives exist at $x \in \Omega$. We call the vector field $\nabla f(x) \in \mathbb{R}^n$ given by

$$\nabla f(x) = \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_n f(x) \end{pmatrix}$$

the *gradient* of f at x.

Note that the directional derivative is related to the gradient via the formula:

$$\partial_v f(x) = \langle \nabla f(x), v \rangle.$$

Example 2.2.8. If $f(x) = |x|$ and $x \neq 0$ then $\nabla f(x) = \begin{pmatrix} \frac{x_1}{|x|} \\ \vdots \\ \frac{x_n}{|x|} \end{pmatrix} = \frac{x}{|x|}$ and $\partial_v f(x) = \langle \nabla f(x), v \rangle$.

Definition 2.2.9. (Partial derivatives for vector-valued functions) Consider a map $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$. We say that the i-th partial derivative of f at $x \in \Omega$ exists, if it exists for all components $f_1, \ldots, f_m$. In that case we write

$$\partial_i f(x) = \begin{pmatrix} \partial_i f_1(x) \\ \vdots \\ \partial_i f_m(x) \end{pmatrix}.$$

The following definition will prove useful when we study differentiable functions.

Definition 2.2.10. (Jacobian matrix) Suppose that $f\colon \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and that all partial derivatives exist at $x \in \Omega$. Then the $(m \times n)$-matrix

$$Df(x) = \begin{pmatrix} \partial_1 f_1(x) & \ldots & \partial_n f_1(x) \\ \vdots & \ddots & \vdots \\ \partial_1 f_m(x) & \ldots & \partial_n f_m(x) \end{pmatrix}$$

is called the *Jacobian matrix* of f in the point x.
If $n = m$ we call $J_f(x) = \det Df(x)$ the *Jacobian determinant* or the *functional determinant* of f at x.

We can write the Jacobian matrix in terms of the gradients of the components:

$$Df(x) = \begin{pmatrix} (\nabla f_1(x))^T \\ \vdots \\ (\nabla f_m(x))^T \end{pmatrix},$$

where the superscript T denotes transposition.

Note for future reference that the Jacobian matrix $Df(x)$ can be consider as a *linear* map from $\mathbb{R}^n$ to $\mathbb{R}^m$: $v \mapsto Df(x)v$.

2.3 Higher partial derivatives

Definition 2.3.1. (Partial derivatives of order k, $C^k(\Omega, \mathbb{R}^n)$) Suppose that we are given a domain $\Omega \subseteq \mathbb{R}^n$.

a) Let $f\colon \Omega \to \mathbb{R}^m$. The *partial derivatives* of order k are inductively defined via

$$\partial_{j_1} \ldots \partial_{j_k} f(x) = \partial_{j_1}(\partial_{j_2} \ldots \partial_{j_k}) f(x) \quad \text{where} \quad j_1, \ldots, j_k \in \{1, \ldots, n\}.$$

Notice that $j_1, \ldots, j_k$ are not necessarily distinct, and that (a priori) their order is important. A common notation is $\frac{\partial^k f(x)}{\partial_{j_1} \ldots \partial_{j_k}}$ or even $\partial_{j_1 j_2 \ldots j_k} f(x)$.

b) Let $C^k(\Omega, \mathbb{R}^m)$ be the set of continuous functions $f\colon \Omega \to \mathbb{R}^m$ whose partial derivatives exist up to order k for all $x \in \Omega$ and are continuous in Ω. If $m = 1$, we write $C^k(\Omega)$. Given a domain $\Theta \subset \mathbb{R}^m$, we will sometimes write $C^k(\Omega, \Theta)$ to be the set of functions $f \in C^k(\Omega, \mathbb{R}^m)$ with $f(\Omega) \subseteq \Theta$.

Proposition 2.3.2. (Exchangeability of partial derivatives, Theorem of Schwarz) *Suppose that $f \in C^2(\Omega)$. Then we have for $1 \leq i, j \leq n$ and for any $x \in \Omega$ that*

$$\partial_i \partial_j f(x) = \partial_j \partial_i f(x)$$

Proof. (Not examinable) Let $\partial_j^t f(x) = \frac{f(x+te_j)-f(x)}{t}$ be the difference quotient of f in x_j. By definition

$$\partial_i \partial_j f(x) = \lim_{s \to 0} (\lim_{t \to 0} \partial_i^s \partial_i^t f(x))$$

We need to show that both limits can be interchanged. By the Intermediate Value Theorem we have for all functions $g\colon \Omega \to \mathbb{R}$, for which $\partial_i g(x)$ exists, that

$$\partial_i^s g(x) = \partial_i g(x + \alpha s e_i) \quad \text{for some } \alpha \in (0,1).$$

If we apply this for $g = \partial_j^t f$ and $g = \partial_i^s f$, we get

$$\begin{aligned} \partial_i^s \partial_j^t f(x) &= \partial_i \partial_j^t f(x + \alpha s e_i) \quad \text{for some } \alpha \in (0,1) \\ &= \partial_j^t (\partial_i f(x + \alpha s e_i)) \\ &= \partial_j (\partial_i f(x + \alpha s e_i + \beta t e_j)) \quad \text{for some } \beta \in (0,1) \end{aligned}$$

Since $\partial_j \partial_i f$ is continuous, it follows that

$$\partial_j \partial_i f(x + \alpha s e_i + \beta t e_j) \xrightarrow{s \to 0} \partial_j \partial_i f(x + \beta t e_j) \xrightarrow{t \to 0} \partial_j \partial_i f(x).$$

□

Corollary 2.3.3. *Suppose that $f \in C^k(\Omega)$. Then all partial derivatives up to order k can be interchanged.*

The following example shows that the condition in Proposition **??** that the second partial derivatives must be continuous is indeed necessary.

Example 2.3.4. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} xy\dfrac{x^2 - y^2}{x^2 + y^2} & \text{for } (x,y) \neq (0,0); \\ 0 & \text{for } (x,y) = (0,0). \end{cases}$$

One can show that $f \in C^1(\mathbb{R}^2)$, but $\partial_x \partial_y f(0,0) = 1$ and $\partial_y \partial_x f(0,0) = -1$.